Laws, the theory matrix, and experiments derived from laws

Memory families compared by proof

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Laws, the theory matrix, and experiments derived from laws



- The second detailed talk: how memory families are compared by laws, and how experiments are derived from the laws' disagreements.

A memory, abstractly

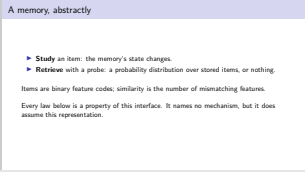
- **Study** an item: the memory's state changes.
- **Retrieve** with a probe: a probability distribution over stored items, or nothing.

Items are binary feature codes; similarity is the number of mismatching features.

Every law below is a property of this interface. It names no mechanism, but it does assume this representation.

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A memory, abstractly



- Laws are stated relative to a starting state and a set of items that may be studied.
- "Mechanism-neutral, not representation-neutral" is an important caveat.

Six laws

Certainty	An item just studied is retrieved by itself with probability 1.
Stability	Studying z leaves the chance that y retrieves itself unchanged.
Order-freedom	Swapping the last two study events leaves retrieval unchanged, for every probe.
Recency	Of two new items studied last, a probe equally similar to both retrieves the later more often.
Stored-only	Nothing is retrieved that was never studied.
Determinacy	Every retrieval has a single outcome.

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Recency	Of two new items studied last, a probe equally similar to both retrieves the later more often.
Stored-only	Nothing is retrieved that was never studied.
Determinacy	Every retrieval has a single outcome.

- Order-freedom is deliberately narrow: only the last two events are swapped, and only retrieval must agree, not the memory state.
- Determinacy is structural as stated, but each "lacks" cell exhibits two outcomes with positive probability, so it fails any probabilistic version too.

Four families

- **Discrimination** (EPAM): a net of feature tests; learning grows a test where codes are confused.
- **Activation** (ACT-R declarative memory): each use adds activation that decays; partial matching; a failure weight. Decay 1, temperature 1: exactly rational.
- **Strength with a criterion** (after DOA): strengths only add; sample by strength, discounted by mismatch; state if above criterion.
- **Global matching** (MINERVA 2): every study stores a trace; a probe activates all traces by the cube of similarity; memory returns the *echo*, their weighted blend.

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- Each family is formalized once, with its parameters free. A published model is a choice of parameters within its family.

The matrix

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
Discrimination	**has**	**has**	lacks	lacks	**has**	**has**
Activation	lacks	lacks	lacks	**has**	**has**	lacks
Strength	lacks	lacks	**has**	lacks	**has**	lacks
Global matching	lacks	lacks	**has**	lacks	lacks	**has**

- **has**: a general proof, for every parameter choice meeting stated conditions.
- lacks: a counterexample, computed exactly on the eight 3-bit codes.
- Every law is had by some family and lacked by another; every family has its own row. A script checks both.

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Strength	lacks	lacks	**has**	lacks	**has**	lacks
Global matching	lacks	lacks	**has**	lacks	lacks	**has**

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- Examples of “has”: EPAM’s certainty needs only that the items be noticeably different; ACT-R’s recency holds for any positive mismatch discount and failure weight (the later item is retrieved exactly twice as often); strength memory and MINERVA are order-free for every parameter choice.
- A “lacks” cell says only that the law fails somewhere. Mapping the parameter regions where a family does have it is open.

The laws were tested, and failed once

- The first four laws were chosen with three families in view.
- Global matching was added as a test. On those four laws, it had *exactly* the row of strength memory.
- So the laws missed something. What is distinctive about global matching? Its output is a blend.
- Study 110, 101, 011; probe with 111 (their prototype, never studied): MINERVA returns 111.
- Hence the stored-only law, which separates them.

Rule: a law enters only with a witness on each side.

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The laws were tested, and failed once

- A law every family satisfies, or satisfies only vacuously, tells theories apart by accident. An earlier all-or-none law, “stable retrieval”, turned out to hold for ACT-R only vacuously, because nothing in ACT-R is retrieved for certain; that is now a theorem.

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Against the evidence

	Certainty	Stability	Order-free	Recency	Stored-only	Determinate
People (judged)	*lacks*	*mixed*	*lacks*	*has*	*mixed*	*mixed*

- A judgment, not a theorem, with sources, kept apart from the proofs.
- Each entry states its **bridge**: how a measure (a report) is taken to show a law (about retrieval).
- Activation fits every settled entry, but each fit rests on unformalized bridges.

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- Sources: Tulving & Pearlstone 1966 (certainty); McGeoch 1932, Barnes & Underwood 1959, Ratcliff, Clark & Shiffrin 1990 (stability: interference in recall, no list-strength effect in recognition); Murdock 1962 and Glanzer & Cunitz 1966 (order, recency); Posner & Keele 1968 and Roediger & McDermott 1995 (stored-only).
- Stored-only is mixed because endorsing a prototype as old doesn't show the prototype was retrieved; next slide.

One experiment, four memories: study, then test

Study 110, 101, 011. Probe each; then the lures 111 (prototype) and 000.

Memory	curve by position	111 called old	111 retrieves itself	000 called old
EPAM	$1, 1, 1$	1	0	1
ACT-R	$\frac{40}{97}, \frac{15}{28}, \frac{120}{157}$	$\frac{55}{61}$	0	$\frac{55}{67}$
Strength	$\frac{8}{17}, \frac{8}{17}, \frac{8}{17}$	1	0	1
MINERVA	$1, 1, 1$	0	1	0

- Derived from laws: EPAM's flat curve (certainty + stability $\Rightarrow$ flat at 1, for any memory).
- Computed: everything else. Recency does not imply a rising self-retrieval curve.
- "Called old" is a report; stored-only is about retrieval. EPAM calls the prototype old by retrieving a studied item.

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EPAM	1, 1, 1	1	0	1
ACT-R	$\frac{66}{91}$, $\frac{66}{91}$, $\frac{126}{91}$	$\frac{66}{91}$	0	$\frac{66}{91}$
Strength	$\frac{1}{3}$, $\frac{2}{3}$, $\frac{3}{3}$	1	0	1
MINERVA	1, 1, 1	0	1	0

- Derived from laws: EPAM's flat curve (certainty + stability ⇒ flat at 1, for any memory).
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- "Called old" is a report; stored-only is about retrieval. EPAM calls the prototype old by retrieving a studied item.

- The table shows why the bridge matters: the two families that satisfy stored-only call the unseen prototype old with certainty, and the one that violates it doesn't, under this crude report rule.

Reversing the workflow

Usually: choose an experiment, compute each theory's predictions.

Instead:

1. find two laws that disagree;
2. derive a design on which they make incompatible predictions;
3. prove the separation, for every memory with either law;
4. state the response model that connects it to what participants say.

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- This makes the matrix an experiment generator rather than a scoreboard.

Recency against order-freedom

Design. Two new items a, b; a probe x equally similar to both. Condition AB studies a then b; condition BA studies b then a; both probe with x.

$$D = P(\text{retrieve } b) - P(\text{retrieve } a), \qquad \Delta = D_{AB} - D_{BA}.$$

Theorems, for any memory:

- order-freedom $\Rightarrow \Delta = 0$;
- recency $\Rightarrow \Delta > 0$;
- so no memory has both laws, wherever such a design exists.

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- *so no memory has both laws, wherever such a design exists.*

- Recency applied to (a, b) gives $D_{AB} > 0$; applied to (b, a) it gives $D_{BA} < 0$. Order-freedom makes the two conditions' retrieval identical.
- These proofs never mention EPAM, ACT-R, or any parameter.

From retrieval to report

Forced choice: report A if a is retrieved, B if b is; otherwise guess B with probability γ.

$$\Delta_{\text{report}} = \Delta + (2\gamma - 1)\,(o_{AB} - o_{BA}),$$

where o is the chance of retrieving neither item.

- Order-freedom: $\Delta_{\text{report}} = 0$, for every γ.
- Recency: $\Delta_{\text{report}} > 0$ with unbiased guessing, or when "neither" is equally likely in both orders.
- A biased guesser can mask or reverse the recency prediction.

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From retrieval to report

- The bridge carries its own condition, stated in the theorem: laws plus a response model give an observable contrast.

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The same design, four memories

Memory	Δ, no prior item	Δ, after 000	why
ACT-R	$10/17$		recency, so $\Delta > 0$
Strength	0	0	order-free
MINERVA	0	0	order-free
EPAM	0	-2	neither law: computed

- ACT-R's reported effect is $10/17$ for every guessing bias: both orders leave the same chance of "neither".
- EPAM shows the most extreme *primacy* possible. After 000, whichever item is learned first is discriminated from it, and the probe always retrieves that one. Its order effects come from the shape of the net, not from time.

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The same design, four memories

- One counterbalanced experiment gives recency, no effect, or primacy, depending on the family, and the matrix says in advance which.

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Lessons, and next

- The durable results are conditional: laws $\Rightarrow$ predictions, families $\Rightarrow$ laws.
- The empirical row will change; when it does, the matrix says what is refuted.
- **Observational semantics** is the main gap: decision and report models, so retrieval laws predict responses.
- **Parameter regions** of the "lacks" cells; **graded laws**; **more families** as tests (REM, connectionist memory, ACT-R with blending).
- **Beyond memory**: strategy choice, attention, production systems.

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Lessons, and next

- Everything in this talk is in the repository: the laws, the cells, the experiments, and a generated matrix with the evidence and its sources.

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